

RAKESH KUMAR

Full-Stack Web Developer / MERN Stack Engineer

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EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering

Graduated: May 2026

Parul University, Vadodara

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP, Web Development.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL

Frontend: React.js (Hooks, Component Architecture), React Router, Redux, Axios, Tailwind CSS, HTML5, CSS3, Bootstrap

Backend & APIs: Node.js, Express.js, REST API Design, JWT Authentication, RBAC, bcrypt, WebSocket

Databases: MongoDB, Mongoose (Schema Design & Validation), MySQL, PostgreSQL, Redis

Tools, Process & Deployment: Git, GitHub, Postman, VS Code, Vercel, Render, npm, Agile

Core CS: Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

PROJECTS

IncidentHub AI — Incident Intelligence Platform | [GitHub](#) | [Demo](#)

TypeScript | React | Node.js | PostgreSQL | Redis | WebSocket | OAuth

- Built a multi-tenant incident intelligence platform with a deterministic correlation engine and an evidence-grounded AI investigation layer, integrating GitHub, Sentry, Slack, and Jira via real OAuth for root-cause analysis.
- Implemented strict multi-tenant RBAC (Owner/Admin/Responder/Viewer), Redis-backed distributed locking, and idempotent HMAC-verified webhooks; validated with 236/236 passing tests including RBAC boundary and cross-tenant isolation suites.
- Designed a citation-backed, versioned postmortem engine and authored complete technical documentation, including an honest limitations section on serverless/WebSocket hosting constraints.

SyncPad — Real-Time Collaborative Code Editor | [GitHub](#) | [Demo](#)

React | TypeScript | Node.js | Yjs (CRDT) | Monaco Editor | WebSocket | Pyodide (WASM)

- Built a real-time collaborative code editor enabling multiple users to edit the same document simultaneously with conflict-free synchronization using Yjs CRDTs, including live multi-cursor presence and awareness.
- Implemented sandboxed, fully in-browser multi-language code execution — JavaScript/TypeScript via Web Workers with timeout protection, and Python via Pyodide (WebAssembly) — removing dependency on third-party execution APIs.
- Designed and deployed a WebSocket-based synchronization server (Node.js + y-websocket) on Render, paired with a React/Vite client on Vercel.

Developer Portfolio Health Checker (PortfolioPulse) | [GitHub](#) | [Demo](#)

React | Vite | Node.js | Express.js | MongoDB | JWT | GitHub REST API

- Built a full-stack career intelligence platform that evaluates GitHub profiles, portfolio websites, and resumes using deterministic, rule-based scoring — every score is backed by explainable evidence, not a black-box output.
- Integrated the GitHub REST API for repository analysis, added a resume/ATS analysis module and recruiter-screening simulation, and built a JWT-authenticated workspace for saved reports and PDF exports.

PROFESSIONAL EXPERIENCE

Software Development Intern — Codetech IT Solutions | Remote

Jan 2026 – Apr 2026

Node.js | Express.js | MongoDB | JWT | RBAC | REST API

- Engineered an Internal Employee Management System (Node.js, Express.js, MongoDB) supporting 3 user roles, replacing manual spreadsheet tracking.
- Secured the platform with JWT-based authentication and layered access control across 3 permission tiers (employee, manager, admin), locking down role-restricted API access.
- Planned and validated RESTful API endpoints supporting full CRUD operations (Create, Read, Update, Delete) across 3 core modules: employee, department, and role.

CERTIFICATIONS & ACHIEVEMENTS

- SQL (Advanced)** — HackerRank
- C++ Programming** — OpenEDG/Cisco
- Introduction to Internet of Things** — NPTEL (IIT Kharagpur), Elite
- AMENTIS: Chartering the Unknown** — Hackathon Participation, IEEE GTBIT Student Branch
- CodeKshetra – Coding Contest** — Participation, GeeksforGeeks (GD Goenka)